MANUAL FOR

KUNDU INTROVERSION EXTRAVERSION INVENTORY

(K. I. E. I.)

By

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The Inventory

The purpose of the inventory is to obtain a reliable measure of introversion-extraversion dimension of adult behaviour or to use it for diagnosis, selection and career guidance. It is developed according to Indian sociocultural pattern. The inventory consists of 70 items with uneven number of response choices divided into 5 blocks. The number of items in each block and the corresponding number of response choices have been shown in Table 1.

Table 1

Number of items and response choices in different blocks

Block	No. of items	No. of response choices
A	11	4
B	17	5
C	14	3
D	14	3
E	14	2

In order to minimise faking effect and also to reduce the nature and number of slanted responses, non-aggressive types of items have been included in the inventory. The mode of response in each block is different and directions for indicating answers for different blocks are given in test booklet.

With a view to check the subjects who have a tendency to respond to the middle most category, from a pattern of systematic presentation, the arrangement of the response choices from 1 to 5 or 1 to 4 or 1 to 3 is not made according to the increasing or decreasing degree of extraversion. To avoid suspicion as to the real purpose of the inventory, the abbreviated name K.I.E.I. (Kundu Introversion-Extraversion Inventory) has been given to it.

The rationale behind dividing the items into five blocks lies with the assumption that specific behaviour which indicate introversion or extraversion dimension of personality pattern may be approached for appraisal from a manifold directions. Since these types of self-reporting inventories are often susceptible to impression management in the form of stereotyped response tendency, it is wise, in our situation, to change the mode of response, particularly when the number of items are large. So far as the contents of the items are concerned, the undernoted features may be noticed in the different blocks :

Block A : Response choices are mostly some objective situations or activities.
Block B : Mostly introspective items.
Block C : Selection of one activity over the other but the activities are mostly of objective nature regarding oneself.
Block D : Mostly objective activities regarding oneself.
Block E : Mostly introspective forced choice items.

Instructions for Administration

The inventory is self-administering in nature. No separate instructions are necessary. In group administration, in order to facilitate starting at a time, it is advantageous to read aloud the instructions appearing in the inventory by the examiner.

No time limit is to be imposed. People usually take less than 30 minutes to complete the inventory.

Accuracy of the results depends on the truthfulness of answers and thorough co-operation of the subjects.

Scoring

The scoring key is prepared on the basis of judgement given by the psychiatrists, psychoanalysts and psychologists. The general order of scoring is such that high score indicates introversion i.e., a negative response is indicative of introversion. But some of the items have been framed in such a way that negative response in these items would indicate extraversion. Classification of items from this point of view is shown in Table 2.

Table 2

Classification of items

Block	Negative answer showing introversion. Item Serial Nos.	Positive snswer showing introversion. Item Serial Nos.
A	1, 4, 6, 10 and 11	2, 3, 5, 7, 8 and 9
B	12, 13, 15, 17, 18, 21, 22, 24, 25, 26 and 28	14, 16, 19, 20, 23 and 27
C	29, 30, 31, 32, 33, 34, 35, 37, 38, 39 and 40	36, 41 and 27
D	43, 44, 45, 46, 49, 50, 51, 52, 53, 54, 55 and 56	47 and 48
E	57, 59, 60, 61, 62, 64, 65, 68 and 69	58, 63, 66, 67 and 70

The different categories of responses are given different weights depending upon the degree of introversion-extraversion they measure. In Table 3 the general principle of scoring is presented.

Table 3

General principle of scoring

Block	Statements indicating introversion for positive responses		Statements indicating extraversion for positive responses	
	Order of response	Order of Score	Order of response	Order of Score
A	1-2-3-4	4-3-2- 1	4-3-2-1	4-3-2-1
B	1-3-5-4-2	5-4-3-2- 1	2-4-5-3-1	5-4-3-2-1
C	1-3 - 2	3 - 2 - 1	2 - 3 - 1	3 - 2 - 1
D	1-3 - 2	3 - 2 - 1	2 - 3 - 1	3 - 2 - 1
E	Yes—No	2 - 1	No—Yes	2 - 1

How to Score

Scoring procedure is simple and may be summarised in the following steps. No scoring key is necessary as the design of the answer sheet has been made with a view to facilitate quick scoring without consulting any key or using any stencil.

1. Count the tick (√) marks in each row in each block and enter the figure under the column T against the respective row.

2. Multiply these totals (T s) in the following order :

Block A : First row total ×1
Second row total ×2
Third row total ×3
Fourth row total ×4

Block B : First row total ×1
Second row total ×2
Third row total ×3
Fourth row total ×4
Fifth row total ×5

Block C, Block D : First row total ×1
Second row total ×2
Third row total ×3

Block E : First row total ×1
Second row total ×2

3. Enter all the products under the column S against the respective row.

4. Find Σ S in each block to obtain the block score and add together all the block scores to obtain total introversion-extraversion score.

5. If block scores are not required, all the entries in the column S may be added together to get the total introversion-extraversion score at a time.

6. Find out the value of C from the Table 5 of this manual and for a ready reference the position of the individual tested may be recorded in the scale given at the bottom of the answer sheet and marked as C.

7. **Check :** Since none of the items are to be omitted in this inventory, a check may be done for omission and wrong counting by adding Ts in different blocks separately. T in each block should be as follows :

Σ T in Block A=11
Σ T in Block B=17
Σ T in Block C=14
Σ T in Block D=14
Σ T in Block E=14

Sample

At present standardisation of K.I.E.I. has been made on sample of the size of 318 adult individuals. The sample has been drawn chiefly from undergrduate degree colleges, teachers' training colleges, Institutes of Business Management and from educated persons holding various services. The minimum educational standard for this sample was B.A./B.Sc. Part-I.

Usually, the minimum standard of education for taking this inventory should be any High Scool leaving examination. But this tentative limit can be either relaxed or enhanced considering the sufficiency or in-sufficiency of English knowledge of the individuals to be tested.

Norms

According to the scoring principle the maximum possible score for this inventory is 241 and the minimum possible score is 70. Within this range, at present norm for the population described above, has been prepared. Norms for bigger and different types of population is expected to be ready very soon. Since there was no significant sex difference, no separate norms for male and female population were determined.

Blockwise and total test norms, in terms of the mean and standard deviation, are given in Table 4.

Table 4

Means and standard deviations

Block	Male		Female		Total	
	Mean	S. D.	Mean	S. D.	Mean	S. D.
A	27.05	3.33	28.3	3.36	27.86	3.36
B	42.96	6.87	41·41	6.36	42.22	6.36
C	27.68	3.99	28.59	4.32	28.91	4.14
D	25.27	3.45	25.70	3.24	25.48	3.30
E	20.91	2.37	21.73	2.40	21.28	2.43
Total test	143.88	12.75	145.36	12.62	144.88	12.70
N	153		165		318	

Meaning of the Scores

Raw scores of the K.I.E.I. have been transformed into C—scale values, the range of each scale-point may be found in Table 5. Just like raw score, low C-scale values indicate extraversion.

Table 5

Range of raw scores against corresponding C-scale points

C-scale point	Raw score range	Classification
0	75 and below	Extremely Extravert
1	76—89	Grossly Extravert
2	90—112	Moderately Extravert
3	113—130	Slightly Extravert
4	131—144	Ambivert
5	145—157	
6	158—171	
7	172—185	Slightly Introvert
8	186—199	Moderately Introvert
9	200—228	Grossly Introvert
10	229 and above	Extremely Introvert

C scores are distributed over eleven interval score points (assuming normal distribution) from 0 through 10 with the population mean fixed at 5 and standard deviation 2. For this inventory, any individual within the limit 4-6 may be considered as neither extravert nor introvert. C scores of 3 and 7 are to be considered as slight deviation from this ambivert condition, 2 and 8 are to be considered as moderate deviation, 1 and 9 are to be considered as gross deviation and 0 and 10 are to be taken as extreme deviation. At this stage, however, no pathological line of demarcation can be suggested. This awaits further study and will be reported in no time.

Reliability

The reliability coefficients are presented in Table 6. Block reliabilities as well as total test reliability have been determined by split-half method applying Spearman-Brown formula. In addition to that reliability has also been computed by Cronbach's α taking each block as a separate subtest.

Table 6

Reliability cofficients (N=318)

Block	Reliability coefficients	
	1st. half vs. 2nd half	Odd vs. even
A	.64	·79
B	.82	.76
C	.60	.86
D	.66	.41
E	.89	.66
Whole test	.82	.71
Whole test by Cronbach's α	.73	

Validity

Validity coefficients, in terms of the block-total correlation, are given in Table 7. The Inventory is also validated against introversion-extraversion score of Eysenck Personality Inventory (E.P.I.). In E.P.I. the increasing order of score corresponds to the increased degree of extravarsion. But in the K.I.E.I. the order is reverse, i.e., the decreasing order of score corresponds to the degree of extraversion. Therefore, coefficient of correlation between K.I.E.I. score and score of E.P.I. is expected to be negative to ensure validity of K.I.E.I. The obtained coefficient of corralation, presented in Table 7, is negative but it is shown without any sign.

Table 7

Validity coefficients

Criteria	Coefficients of correlation
Block A	.41
Block B	.77
Block C	.42
Block D	.55
Block E	.49
E. P. I.	.47

Intercorrelations

In Table 8 are reported the coefficients of intercorrelation of the five blocks of the inventory. These inter-block coefftcients of correlation tentatively indicate that the blocks are more or less factorially independent of each justifying apparently the subgrouping of the items.

Table 8

Cofficients of inter-block correlation

	A	B	C	D	E
A	( )	.15	.17	.15	.11
B	.15	( )	.13	.24	.18
C	.17	.13	( )	.01	.22
D	.15	.24	.01	( )	.23
E	.11	.18	.22	.23	( )

Communication

Further studies with K.I.E.I. are in progress. The author will be glad to receive information regarding the use of the inventory. Communications should be addressed to :

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